Exercises week 4 – Answers

8CM10 - Introduction to Quantum Physics and Structure of Matter

Atoms

1. Electrons in atoms are known to have kinetic energies in the range of a few eV. Show that the uncertainty principle allows electrons of this energy to be confined in a region the size of an atom (0.1 nm)

$$\Delta p \sim \frac{\hbar}{\Delta x} = \frac{1}{c}\frac{\hbar c}{\Delta x} = \frac{1}{c}\frac{197\,\text{eV}\cdot\text{nm}}{0.10\,\text{nm}} = 1970\,\text{eV}/c$$

$$K_{\text{av}} = \frac{p_{\text{av}}^2}{2m} = \frac{p_{\text{av}}^2 c^2}{2mc^2} = \frac{(1970\,\text{eV})^2}{2(0.511\times10^6\,\text{eV})} = 3.8\,\text{eV}$$

This energy is consistent with the observed energies of electrons in atoms.

Bohr Model

2. In the $n = 3$ state of hydrogen, find the electron's velocity, kinetic energy, and potential energy.

$$r_3 = 9a_0 = 9(0.0529\text{ nm}) = 0.476\text{ nm}$$

$$v = \frac{n\hbar}{mr} = c\frac{n\hbar c}{mc^2 r} = c\frac{3(1240\,\text{eV}\cdot\text{nm})/2\pi}{(0.511\times10^6\,\text{eV})(0.476\,\text{nm})} = 2.43\times10^{-3}c = 7.30\times10^5\,\text{m/s}$$

$$U = -\frac{e^2}{4\pi\varepsilon_0}\frac{1}{r} = -\frac{1.440\,\text{eV}\cdot\text{nm}}{0.476\text{ nm}} = -3.02\,\text{eV}$$

$$K = \frac{e^2}{8\pi\varepsilon_0}\frac{1}{r} = \frac{1.440\,\text{eV}\cdot\text{nm}}{2(0.476\text{ nm})} = 1.51\,\text{eV}$$

3. An electron is in the $n = 5$ state of hydrogen. To what states can the electron make transitions, and what are the energies of the emitted radiations?

The energy of the initial $n = 5$ state is $E_5 = \dfrac{-13.6\text{ eV}}{25} = -0.544\text{ eV}$. An electron in this state can make transitions to any of the lower states with $n = 4$ ($E_4 = -0.850$ eV), $n = 3$ ($E_3 = -1.51$ eV), $n = 2$ ($E_2 = -3.40$ eV), and $n = 1$ ($E_1 = -13.6$ eV). The transition energies are:

$$5 \to 4: \quad \Delta E = E_5 - E_4 = -0.544\text{ eV} - (-0.850\text{ eV}) = 0.306\text{ eV}$$
$$5 \to 3: \quad \Delta E = E_5 - E_3 = -0.544\text{ eV} - (-1.51\text{ eV}) = 0.97\text{ eV}$$
$$5 \to 2: \quad \Delta E = E_5 - E_2 = -0.544\text{ eV} - (-3.40\text{ eV}) = 2.86\text{ eV}$$
$$5 \to 1: \quad \Delta E = E_5 - E_1 = -0.544\text{ eV} - (-13.6\text{ eV}) = 13.1\text{ eV}$$

4. Use the Bohr formula to find the energy differences $E(n_1 \to n_2) = E_{n_1} - E_{n_2}$ and show that:
   a. $E(4 \to 2) = E(4 \to 3) + E(3 \to 2)$
   b. $E(4 \to 1) = E(4 \to 2) + E(2 \to 1)$

(a) $$E(4 \to 2) = E_4 - E_2 = (-13.6\text{ eV})\left(\frac{1}{4^2} - \frac{1}{2^2}\right) = 2.55\text{ eV}$$

$$E(4 \to 3) = E_4 - E_3 = (-13.6\text{ eV})\left(\frac{1}{4^2} - \frac{1}{3^2}\right) = 0.661\text{ eV}$$

$$E(3 \to 2) = E_3 - E_2 = (-13.6\text{ eV})\left(\frac{1}{3^2} - \frac{1}{2^2}\right) = 1.89\text{ eV}$$

$$E(4 \to 3) + E(3 \to 2) = 0.661\text{ eV} + 1.89\text{ eV} = 2.55\text{ eV} = E(4 \to 2)$$

(b) $$E(4 \to 1) = E_4 - E_2 = (-13.6\text{ eV})\left(\frac{1}{4^2} - \frac{1}{1^2}\right) = 12.8\text{ eV}$$

$$E(2 \to 1) = E_2 - E_1 = (-13.6\text{ eV})\left(\frac{1}{2^2} - \frac{1}{1^2}\right) = 10.2\text{ eV}$$

$$E(4 \to 2) + E(2 \to 1) = 2.55\text{ eV} + 10.2\text{ eV} = 12.8\text{ eV} = E(4 \to 1)$$

5. Find the shortest and the longest wavelengths of the Lyman series of singly ionized helium.

The Lyman series consists of transitions that end in the $n = 1$ level. The smallest energy difference, corresponding to the longest wavelength, is $n = 2$ to $n = 1$.

$$\Delta E = E_2 - E_1 = (-13.6\text{ eV})2^2\left(\frac{1}{2^2} - \frac{1}{1^2}\right) = 40.8\text{ eV}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{1240\text{ eV}\cdot\text{nm}}{40.8\text{ eV}} = 30.4\text{ nm}$$

The largest energy difference would correspond to transitions from $n = \infty$ to $n = 1$:

$$\Delta E = E_\infty - E_1 = (-13.6\text{ eV})2^2\left(0 - \frac{1}{1^2}\right) = 54.4\text{ eV}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{1240\text{ eV}\cdot\text{nm}}{54.4\text{ eV}} = 22.8\text{ nm}$$

6. Draw an energy-level diagram showing the lowest four levels of singly ionized helium. Show all possible transitions from the levels and label each transition with its wavelength.

Using Equation 6.38, we have $E_n = (-13.6\text{ eV})Z^2/n^2 = (-54.4\text{ eV})/n^2$, so $E_1 = -54.40$ eV, $E_2 = -13.60$ eV, $E_3 = -6.04$ eV, $E_4 = -3.40$ eV. The possible transitions are:

$4 \to 1$: $\Delta E = E_4 - E_1 = 51.00$ eV $\quad \lambda = hc/\Delta E = 24.31$ nm

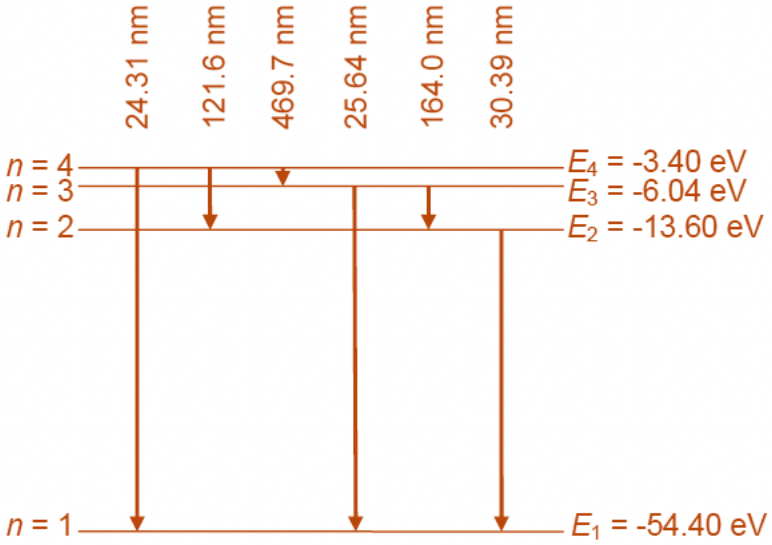
$4 \to 2$: $\Delta E = E_4 - E_2 = 10.20$ eV $\quad \lambda = hc/\Delta E = 121.6$ nm

$4 \to 3$: $\Delta E = E_4 - E_3 = 2.64$ eV $\quad \lambda = hc/\Delta E = 469.7$ nm

$3 \to 1$: $\Delta E = E_3 - E_1 = 48.36$ eV $\quad \lambda = hc/\Delta E = 25.64$ nm

$3 \to 2$: $\Delta E = E_3 - E_2 = 7.56$ eV $\quad \lambda = hc/\Delta E = 164.0$ nm

$2 \to 1$: $\Delta E = E_2 - E_1 = 40.80$ eV $\quad \lambda = hc/\Delta E = 30.39$ nm



The Franck-Hertz Experiment

7. A hypothetical atom has only two excited states, at 4.0 and 7.0 eV, and has a ground-state ionization energy of 9.0 eV. If we used a vapor of such atoms for the Franck-Hertz experiment, for what voltages would we expect to see decreases in the current? List all voltages up to 20 V.

Let $V_1 = 4$ V, $V_2 = 7$ V, and $V_3 = 9$ V. Then decreases in the current should be observed at the following voltages:

$V_1 = 4$ V
$V_2 = 7$ V
$2V_1 = 8$ V
$V_3 = 9$ V
$V_1 + V_2 = 11$ V
$3V_1 = 12$ V
$V_1 + V_3 = 13$ V
$2V_2 = 14$ V
$2V_1 + V_2 = 15$ V
$4V_1 = 16$ V
$2V_1 + V_3 = 17$ V
$2V_3 = 18$ V
$3V_1 + V_3 = 19$ V
$5V_1 = 20$ V